ECON 0100 | HW C2 | Taxes + Subsidies

Due: Sunday, Oct. 18

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | The Butterbeer Tax

The Ministry of Magic was in some financial trouble after their considerable expenditures during the war. They decided to impose a tax of 6 galleons on the sale of butterbeer as a source of funding. The supply and demand curves for butterbeer can be represented by the following equations.

$$D : P_B = 100 - \frac{1}{2}Q_D \qquad S : P_S = 20 + \frac{1}{2}Q_S$$

Use a graph and algebra to illustrate what happened to the market before and after the tax. Start with the pre-tax equilibrium. Then use the tax equation ($P_B = \tau + P_S$) to solve for equilibrium quantity and the two prices. Conclude with the remaining relevant components.

Pre-Tax Price ($\hat{P}$):	_____	Pre-Tax Quantity ($\hat{Q}$):	_____
Post-Tax Quantity (Q^τ):	_____		
Buyer Price (P_B):	_____	Seller Price (P_S):	_____
Government Revenue:	_____	Deadweight Loss (DWL):	_____
Tax Incidence on Buyers:	_____	Tax Incidence on Sellers:	_____

Q2 | The Butterbeer Subsidy

Now suppose that instead of the tax, the Ministry of Magic pays a 6 galleon subsidy on every butterbeer sold. Use a graph and algebra to discuss the impact of the subsidy on the market. Use the subsidy equation ($P_S = s + P_B$) to solve for equilibrium quantity, the two prices, and DWL.

Post-Subs Quantity (Q^{Sub}): _____ Deadweight Loss: _____

Buyer Price (P_B): _____ Seller Price (P_S): _____

For the bottles sold beyond the pre-subsidy quantity, how does marginal benefit compare with marginal cost?

a) Marginal benefit is above marginal cost, so these added trades raise total surplus.

b) Marginal benefit is below marginal cost, so these added trades reduce total surplus once the Ministry's spending is counted.

c) Marginal benefit equals marginal cost, so these added trades leave total surplus unchanged.

d) The subsidy lowers the sellers' marginal cost, so these added trades cost society nothing.